

The Alternating Subset Expansion of Iterated Finite Differences

Abstract

We prove that an iterated finite difference of an arbitrary function can be written as an alternating sum over all subsets of the applied shifts. The argument uses only elementary properties of finite sums, a binary decomposition of subsets, and mathematical induction. To handle repeated shift values correctly, the sum is taken over subsets of the set of *positions* in an ordered family, rather than over a set in which equal shift values would be identified.

1 Setting and notation

Let $(A, +, 0)$ be a commutative additive monoid, let K be a commutative ring, and let

$$f: A \longrightarrow K$$

be an arbitrary function. No regularity, continuity, differentiability, or linearity assumption is imposed on f . The identity proved below is a finite algebraic and combinatorial identity.

For every shift $h \in A$, define the one-step finite-difference operator by

$$(\Delta_h f)(D) = f(D) - f(D + h), \quad D \in A. \quad (1)$$

Thus our sign convention is “the value at the current point minus the value at the shifted point.”

Given a finite ordered family of shifts

$$\mathbf{y} = (y_1, y_2, \dots, y_n),$$

define its iterated finite difference by

$$\Delta_{\mathbf{y}} f := \Delta_{y_1} (\Delta_{y_2} (\dots (\Delta_{y_n} f))). \quad (2)$$

When $n = 0$, we use the convention $\Delta_{\emptyset} f = f$.

Write

$$[n] := \{1, 2, \dots, n\}, \quad y_V := \sum_{a \in V} y_a \quad (V \subseteq [n]),$$

with the empty-sum convention $y_{\emptyset} = 0$. Since addition in A is associative and commutative, y_V does not depend on the order used to sum the elements indexed by V .

It is important that V is a subset of the *index set*. For example, if $y_1 = y_2$, the choices $V = \{1\}$ and $V = \{2\}$ remain distinct and must produce two separately counted terms. This is exactly the multiplicity required by repeated application of finite-difference operators.

2 The main identity

Theorem 2.1 (Alternating subset expansion). *For every $n \in \mathbb{N}$, every ordered family $(y_1, \dots, y_n)$ of elements of A , every function $f: A \rightarrow K$, and every $D \in A$, one has*

$$\boxed{(\Delta_{y_1} \Delta_{y_2} \dots \Delta_{y_n} f)(D) = \sum_{V \subseteq [n]} (-1)^{|V|} f\left(D + \sum_{a \in V} y_a\right)}. \quad (3)$$

The composition of operators on the left is understood according to (2).

Every selected shift contributes one minus sign. Consequently, a subset V containing $|V|$ selected positions has coefficient $(-1)^{|V|}$.

3 Two elementary facts about finite sums

The proof separates paired terms into two sums and moves a pointwise minus sign outside a finite sum. We first record these two operations in purely mathematical form.

Lemma 3.1 (Splitting a sum of pairs). *Let S be a finite set and let $F, G: S \rightarrow K$. Then*

$$\sum_{s \in S} (F(s) + G(s)) = \sum_{s \in S} F(s) + \sum_{s \in S} G(s). \quad (4)$$

Proof. We argue by induction on $|S|$. If $S = \emptyset$, both sides are zero. Otherwise, choose $s_0 \in S$ and write the disjoint union $S = \{s_0\} \sqcup S'$. The induction hypothesis gives

$$\begin{aligned} \sum_{s \in S} (F(s) + G(s)) &= F(s_0) + G(s_0) + \sum_{s \in S'} (F(s) + G(s)) \\ &= F(s_0) + G(s_0) + \sum_{s \in S'} F(s) + \sum_{s \in S'} G(s) \\ &= \left(F(s_0) + \sum_{s \in S'} F(s) \right) + \left(G(s_0) + \sum_{s \in S'} G(s) \right). \end{aligned}$$

In the last step, we used associativity and commutativity of addition in K . The final expression is the right-hand side of (4). $\square$

Lemma 3.2 (Negation of a finite sum). *Let S be a finite set and let $F: S \rightarrow K$. Then*

$$\sum_{s \in S} (-F(s)) = - \sum_{s \in S} F(s). \quad (5)$$

Proof. Set $T = \sum_{s \in S} F(s)$. Adding the two pointwise sums gives

$$T + \sum_{s \in S} (-F(s)) = \sum_{s \in S} (F(s) - F(s)) = 0.$$

The second summand on the left is therefore the additive inverse of T , which proves (5). $\square$

Together, the two lemmas give the familiar identity

$$\sum_{s \in S} F(s) - \sum_{s \in S} G(s) = \sum_{s \in S} (F(s) - G(s)). \quad (6)$$

4 The binary decomposition of subsets

The combinatorial core of the induction step is as follows. Start with the family

$$\mathbf{y} = (y_1, \dots, y_n)$$

and place one new shift h in front of it, producing

$$(h, \mathbf{y}) = (h, y_1, \dots, y_n).$$

Every selection of positions in the new family belongs to exactly one of the following two mutually exclusive classes:

1. h is not selected, and all remaining choices are uniquely described by a subset $V \subseteq [n]$;

2. h is selected, and the remaining choices are again uniquely described by a subset $V \subseteq [n]$.

Thus all selections from the new family are in bijection with the disjoint union

$$\{(0, V) : V \subseteq [n]\} \sqcup \{(1, V) : V \subseteq [n]\},$$

where the first coordinate records whether h is selected.

For a fixed $V \subseteq [n]$, if h is not selected, the number and sum of the selected shifts are

$$|V|, \quad y_V.$$

If h is selected, they become

$$|V| + 1, \quad h + y_V.$$

The corresponding sign changes according to

$$(-1)^{|V|+1} = (-1)^{|V|}(-1) = -(-1)^{|V|}. \quad (7)$$

Hence every term containing h is the negative of the corresponding old term evaluated after shifting the base point by h .

For a fully indexed description, set $z_0 = h$ and $z_i = y_i$ for $1 \leq i \leq n$. The alternating subset sum for the new family is then

$$\begin{aligned} & \sum_{U \subseteq \{0,1,\dots,n\}} (-1)^{|U|} f\left(D + \sum_{a \in U} z_a\right) \\ &= \sum_{V \subseteq [n]} (-1)^{|V|} f(D + y_V) + \sum_{V \subseteq [n]} (-1)^{|V|+1} f(D + h + y_V) \\ &= \sum_{V \subseteq [n]} (-1)^{|V|} f(D + y_V) - \sum_{V \subseteq [n]} (-1)^{|V|} f(D + h + y_V). \end{aligned} \quad (8)$$

Here every U is a subset of labelled positions, so repeated shift values are not identified.

5 Detailed inductive proof of the main theorem

Proof of Theorem 2.1. We proceed by induction on the number n of shifts.

Step 1: The base case.

Suppose $n = 0$. There is no finite-difference operator to apply, so by convention the left-hand side is

$$(\Delta_{\emptyset} f)(D) = f(D).$$

On the right-hand side, the power set of the empty index set has exactly one element, namely the empty set. Moreover, $|\emptyset| = 0$ and $y_{\emptyset} = 0$. Therefore

$$\sum_{V \subseteq [0]} (-1)^{|V|} f(D + y_V) = (-1)^0 f(D + 0) = f(D).$$

The two sides agree, proving the base case.

Step 2: The induction hypothesis.

Fix a family $\mathbf{y} = (y_1, \dots, y_n)$ of length n . Assume that for every $X \in A$,

$$(\Delta_{\mathbf{y}} f)(X) = \sum_{V \subseteq [n]} (-1)^{|V|} f(X + y_V). \quad (9)$$

The phrase “for every X ” is essential: the induction hypothesis will be used both at $X = D$ and at $X = D + h$.

Step 3: Expand the outermost finite difference.

Place an arbitrary new shift $h \in A$ in front of the old family. By the definition (1),

$$(\Delta_h \Delta_y f)(D) = (\Delta_y f)(D) - (\Delta_y f)(D + h). \quad (10)$$

Apply the induction hypothesis (9) first with $X = D$ and then with $X = D + h$. This yields

$$\begin{aligned} (\Delta_h \Delta_y f)(D) &= \sum_{V \subseteq [n]} (-1)^{|V|} f(D + y_V) \\ &\quad - \sum_{V \subseteq [n]} (-1)^{|V|} f((D + h) + y_V). \end{aligned} \quad (11)$$

By associativity of addition, $(D + h) + y_V = D + (h + y_V)$, so the last expression can be written as

$$\sum_{V \subseteq [n]} (-1)^{|V|} f(D + y_V) - \sum_{V \subseteq [n]} (-1)^{|V|} f(D + h + y_V). \quad (12)$$

Step 4: Identify the complete subset sum for the new family.

By the binary decomposition of the preceding section, selections not containing h contribute the first sum in (12). Selections containing h have one additional selected element, and therefore one additional minus sign; they contribute the negative of the second sum. More explicitly, using (7), expression (12) becomes

$$\begin{aligned} &\sum_{V \subseteq [n]} (-1)^{|V|} f(D + y_V) + \sum_{V \subseteq [n]} (-1)^{|V|+1} f(D + h + y_V) \\ &= \sum_{U \subseteq \{0,1,\dots,n\}} (-1)^{|U|} f\left(D + \sum_{a \in U} z_a\right). \end{aligned}$$

The right-hand side is precisely the alternating subset expansion for the new family $(h, y_1, \dots, y_n)$ of length $n + 1$. Thus the assertion for length n implies the assertion for length $n + 1$.

Combining the base case with the induction step proves (3) for every $n \in \mathbb{N}$. $\square$

6 Direct checks in low degrees

The first few cases make the origin and sign of every term transparent.

One shift

For the single shift y_1 , the two index subsets are $\emptyset$ and $\{1\}$. Hence

$$(\Delta_{y_1} f)(D) = f(D) - f(D + y_1).$$

Two shifts

Expanding one layer at a time gives

$$\begin{aligned} (\Delta_{y_1} \Delta_{y_2} f)(D) &= (\Delta_{y_2} f)(D) - (\Delta_{y_2} f)(D + y_1) \\ &= f(D) - f(D + y_2) - f(D + y_1) + f(D + y_1 + y_2). \end{aligned}$$

These four terms correspond respectively to

$$\emptyset, \quad \{2\}, \quad \{1\}, \quad \{1, 2\}.$$

Subsets of even cardinality have a positive sign, while subsets of odd cardinality have a negative sign.

Three shifts

For three positions there are $2^3 = 8$ subsets, and the formula reads

$$\begin{aligned} (\Delta_{y_1} \Delta_{y_2} \Delta_{y_3} f)(D) = & f(D) - f(D + y_1) - f(D + y_2) - f(D + y_3) \\ & + f(D + y_1 + y_2) + f(D + y_1 + y_3) + f(D + y_2 + y_3) \\ & - f(D + y_1 + y_2 + y_3). \end{aligned}$$

The numbers of terms in the four layers are $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3}$, and their signs alternate. This is the general subset formula grouped according to cardinality.